

MEKONG CLI — Hướng Dẫn Sử Dụng Hiệu Quả

Version 5.0.0 | AI-Operated Business Platform | MIT License "Speed is the essence of war." — Sun Tzu

1. TỔNG QUAN

Mekong CLI là nền tảng AI tự động hóa doanh nghiệp, biến mục tiêu ngôn ngữ tự nhiên thành hành động thực thi qua pipeline **Plan → Execute → Verify (PEV)**.

Thống kê: 289 commands • 245 skills • 176 contracts • 13 agents • 9 sub-apps

Cách hoạt động

```
"Tạo FastAPI service với JWT auth"
↓ PLAN → Phân tách thành các bước (LLM-powered)
↓ EXECUTE → Thực thi từng bước (shell/llm/tool/browse/evolve)
↓ VERIFY → Kiểm tra kết quả theo quality gates
↓ DONE → Hoàn thành, MCU credits trừ
```

2. CÀI ĐẶT & KHỞI ĐỘNG

2.1 Cài đặt

```
pip install mekong-cli
```

2.2 Cấu hình LLM (3 biến môi trường)

```
export LLM_BASE_URL="https://api.openai.com/v1"
export LLM_API_KEY="sk-..."
export LLM_MODEL="gpt-4o"
```

2.3 Chạy lệnh đầu tiên

```
mekong cook "Tạo Python calculator với tests"
```

2.4 LLM Fallback Chain (9 tầng)

Ưu tiên	Provider	Env Var
0	Primary (user config)	LLM_BASE_URL + LLM_API_KEY
1	OpenRouter	OPENROUTER_API_KEY
2	AgentRouter	AGENTROUTER_API_KEY
3	Qwen/DashScope	DASHSCOPE_API_KEY

4	DeepSeek	DEEPSEEK_API_KEY
5	Anthropic	ANTHROPIC_API_KEY
6	OpenAI	OPENAI_API_KEY
7	Google Gemini	GOOGLE_API_KEY
8	Ollama (local)	OLLAMA_BASE_URL
9	Offline	(luôn có sẵn)

3. KIẾN TRÚC 5 LỚP

Lớp	Emoji	Chức năng	Ví dụ Commands
Founder	👑	Chiến lược	/annual /okr /fundraise /swot /pitch /vc/* /ipo/*
Business	🏢	Vận hành	/sales /marketing /finance /hr /pricing /brand
Product	📦	Sản phẩm	/plan /sprint /roadmap /brainstorm /scope
Engineer	⚙️	Kỹ thuật	/cook /fix /code /test /deploy /review
Ops	🔧	Vận hành hạ tầng	/audit /health /security /status /clean

4. TOP 10 COMMANDS QUAN TRỌNG

Command	Chức năng	MCU
mekong cook	Full PEV pipeline - thực thi mọi task	3
mekong fix	Debug & fix bugs tự động	3
mekong plan	Tạo implementation plan	1
mekong review	Code quality review	1
mekong deploy	Deployment orchestration	5
mekong company/init	Khởi tạo agentic company	3
mekong founder/validate	Customer discovery & PMF	3
mekong founder/pitch	Pitch deck practice	3
mekong founder/vc/cap-table	Dilution modeling	5
mekong founder/ipo/ipo-day	IPO Day execution	5

CLI Flags

Flag	Tác dụng
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<code>--verbose / -v</code>	Hiển thị chi tiết từng bước
<code>--dry-run / -n</code>	Chỉ plan, không thực thi
<code>--strict</code>	Dừng ngay khi gặp lỗi verification
<code>--json / -j</code>	Output JSON cho máy đọc

5. COMMANDS CHI TIẾT THEO LỚP

5.1 Founder Layer 🏰

VC Suite (5 modules):

- `vc/bootstrap` — Bootstrap funding
- `vc/cap-table` — Cap table & dilution modeling
- `vc/map` — VC landscape mapping
- `vc/negotiate` — Term negotiation
- `vc/term-sheet` — Term sheet drafting

IPO Suite (8 modules):

- `ipo/pre-ipo` — Chuẩn bị Pre-IPO
- `ipo/s1` — S-1 filing
- `ipo/roadshow` — IPO roadshow planning
- `ipo/ipo-day` — IPO Day execution
- `ipo/public-co` — Post-IPO governance
- `ipo/insider` — Insider trading compliance
- `ipo/succession` — CEO succession planning

5.2 Business Layer 🏢

- `/sales` — Pipeline management, outreach, lead qualify
- `/marketing` — Campaigns, content, ads, SEO, CRO
- `/finance` — P&L, budgets, financial reporting
- `/hr` — Hiring, team building
- `/client` — Client onboarding, contracts, invoices

5.3 Product Layer 📦

- `/plan` — Implementation planning (modes: hard/fast/standard)
- `/sprint` — Sprint management
- `/roadmap` — Product roadmap
- `/brainstorm` — Ideation sessions
- `/idea` — Full company architecture from idea (25 steps, Zero→IPO)

5.4 Engineer Layer ⚙️

- `/cook` — Full PEV pipeline (modes: auto/fast/parallel/interactive)
- `/fix` — Debug & auto-fix
- `/code` — Code analysis, TDD, optimization
- `/test` — Test suite execution
- `/deploy` — Deployment orchestration
- `/review` — Code review

- `/dev` — Audit, debug, scaffold, refactor

5.5 Ops Layer 🔧

- `/audit` — System audit
- `/health` — Health sweep
- `/security` — Security checks
- `/status` — System status
- `/daily` — Daily status report
- `/ship` — Ship code: lint → test → commit → push → deploy

6. PEV ENGINE (Core)

6.1 Orchestrator (1023 lines)

```
RecipeOrchestrator
├─ StepExecutor      — Thực thi + verify từng bước
├─ RollbackHandler  — Rollback khi thất bại
└─ ReportFormatter  — Rich terminal reporting
```

6.2 Planner (660 lines)

- Phân tách goal thành atomic tasks qua LLM
- DAG-aware planning (parallel khi có thể)
- Re-planning tự động khi branch thất bại
- Agent keyword routing (git/file/shell/lead/content/browse/evolve)

6.3 Executor (490 lines) — 5 chế độ

Mode	Mô tả
shell	subprocess.run với retry + sanitization
llm	LLM generation qua provider
api	HTTP request (GET/POST/PUT/DELETE)
tool	ToolRegistry execution
browse	BrowserAgent (web extraction)

6.4 Verifier (483 lines)

- Exit code check
- File exists/not exists
- Output contains/not contains
- Custom check (shell command)
- **Bình Pháp Quality Gates:** 0 TODO, 0 console.log, 0 `:any`, 0 vulnerabilities

6.5 Self-Healing Flow

```
Step fails → Reflection Engine phân tích root cause
└─ World Model kiểm tra environment
└─ LLM tạo correction strategy
```

- ↓ Retry với exponential backoff (max 3 retries)
- ↓ Nếu vẫn fail → Rollback + Dead Letter Queue

7. HỆ THỐNG 13 AGENTS

Agent	Chức năng
GitAgent	Git operations (commit, branch, merge)
FileAgent	File CRUD operations
ShellAgent	Shell command execution
RecipeCrawler	Recipe file discovery
LeadHunter	Sales lead generation
ContentWriter	Content/blog/SEO writing
DatabaseAgent	Database operations
MonitorAgent	System resource monitoring
PluginAgent	Plugin management
WorkspaceAgent	Workspace management
NetworkAgent	Network operations
AGIBridge	AGI subsystem bridge

Tất cả agents kế thừa từ AgentBase và tuân theo flow: **plan** → **execute** → **verify**.

8. AGI v2 — 9 HỆ THỐNG CON

#	Hệ thống	Chức năng
1	NLU	Intent classification, command parsing
2	Memory	Persistent execution memory
3	Reflection	Past failure analysis
4	WorldModel	Environment state tracking
5	ToolRegistry	Dynamic tool management
6	BrowserAgent	Web content extraction
7	Collaboration	Multi-agent coordination
8	CodeEvolution	Code quality evolution
9	VectorMemory	Semantic vector search

AGI Score Engine (0-100)

Dimension	Weight
Module availability	45%
Cross-wiring depth	25%
Runtime metrics	15%
Self-improvement	15%

Grade: S (95+), A (85+), B (75+), C (65+), D (50+), F (<50)

9. WORKFLOWS (Slash Commands)

Chiến lược & Doanh nghiệp

Command	Mô tả
/idea	Tạo full company architecture từ ý tưởng
/binh-phap	Strategic execution framework
/studio	Strategy, portfolio, bootstrap, invest
/business	Client onboard, revenue engine
/daily	Daily status report
/win	WIN-WIN-WIN validation

Kỹ thuật & Phát triển

Command	Mô tả
/cook	Intelligent feature implementation
/plan	Research and analysis planning
/code	TDD, code analysis, optimization
/dev	Audit, debug, deploy, design
/ship	Lint → test → commit → push → deploy
/git	Commit, PR, rebase, worktrees
/build	Code generation + tests + CI + deploy

Sales & Marketing

Command	Mô tả
/sales	Pipeline, outreach, deal close

/marketing	Campaigns, content, ads, SEO
/content	Article, SEO, social posts
/money	Lead generation + content + outreach

Operations

Command	Mô tả
/ops	Health sweep, sprints, budgets
/cto	Architecture, deploy, incident response
/approve	Approve pending content/decisions
/quick-start	New project → production in 5 steps

10. PLUGIN ECOSYSTEM

Cài đặt Plugin

```
mekong plugin install <name>
```

Plugin Types

- **agent** — Agent mở rộng
- **provider** — LLM provider mới
- **hook** — Hook vào pipeline
- **recipe** — Recipe templates

Marketplace API

- Search, sort by: popularity/downloads/rating
- Featured & trending discovery
- Rating system

11. TELEGRAM COMMANDER BOT

Commands

Command	Chức năng
/cook <goal>	Queue task
/spawn <project> <goal>	Task cho project cụ thể
/tasks	Xem pending tasks
/sessions	List active terminals
/status	System health

/schedule	Scheduled jobs
/agi start	Bắt đầu AGI self-improvement

Cấu hình

```
export MEKONG_TELEGRAM_TOKEN="bot-token-here"
```

File config: `.mekong/telegram.yaml`

12. SWARM REGISTRY (Multi-Node)

Phân phối task across multiple Mekong gateways:

```
# Register node
registry.register_node(name, host, port, token)

# Health check all nodes
registry.check_all_health()

# Dispatch goal to specific node
registry.dispatch_goal(node_id, goal)
```

Priority: Healthy remote node → Local agent fallback

13. EVENTBUS & TELEMETRY

40+ Event Types (7 categories)

- **Pipeline:** plan/execute/verify start/done
- **Step:** start/done/fail/retry/rollback
- **Recipe:** load/save/auto_save
- **Agent:** register/invoke/response
- **System:** health/error/shutdown
- **AGI:** score/self_heal/evolve
- **Billing:** recorded/overage/period_closed

Tiered Telemetry

Tier	Retention	Content
T0 (Hot)	24h	Real-time metrics
T1 (Warm)	30 days	Aggregated stats
T2 (Cold)	Vĩnh viễn	Full audit trail

14. DEPLOY (4-Layer, \$0 Cost)

Layer	Platform	Cost
Frontend	Cloudflare Pages	\$0
Edge API	Cloudflare Workers	\$0
App	Vercel / CF Pages SSR	\$0
Backend	Fly.io	\$0-20

```
bash mekong/infra/scaffold.sh myproject startup # frontend + API
bash mekong/infra/scaffold.sh myproject scale # all 4 layers
```

⚠ **BANNED:** `vercel --prod` — chỉ deploy qua `git push`

15. BÌNH PHÁP QUALITY FRAMEWORK ()

Front	Tiêu chuẩn
Tech Debt	0 TODO/FIXME, 0 console.log
Type Safety	0 any types, strict mode
Performance	Build < 10s, LCP < 2.5s
Security	Input validation, no secrets
UX	Loading states, error boundaries
Documentation	Self-documenting code

16. MCU BILLING

Tier	Credits/tháng	Giá
Starter	200	\$49
Pro	1,000	\$149
Enterprise	Unlimited	\$499

- 1 MCU = 1 credit, trừ sau khi delivery thành công
- Zero balance → HTTP 402
- Polar.sh webhooks = payment source chính

17. SUB-APPLICATIONS (9 Apps)

App	Chức năng
openclaw-worker	Tôm Hùm autonomous daemon

raas-gateway	RaaS API gateway
algo-trader	Algorithm trading bot
engine	Core execution engine
vibe-coding-cafe	Vibe coding app
docs	Documentation site

18. BẢO MẬT

Dangerous Patterns (tự động chặn)

```
rm -rf / , mkfs , dd if= , chmod -R 777 / , curl|sh , shutdown , reboot
```

Auth Stack

- JWT authentication
- Session management
- Tenant-scoped auth
- Permission registry
- Device certificates
- Machine fingerprinting

19. QUICK REFERENCE

```
# Khởi tạo
pip install mekong-cli
mekong doctor check

# Sử dụng hàng ngày
mekong cook "mô tả task"           # Thực thi task
mekong plan "feature mới"           # Lập kế hoạch
mekong fix "bug description"         # Fix bug
mekong deploy                       # Deploy
mekong daily                         # Báo cáo ngày

# Kiểm tra
mekong health                       # Health check
mekong status                       # System status
mekong help                         # Danh sách commands
```